Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
F		**546567**

TITLE

Summary

BHSBVHBHSBVSDBJVBJDSBJVBSDJBVJDBBVJBDJVBJDB

Keywords: SDCBJHDSBJCDSJBJCDBSJVBSDB;SDCHJSBVSD;SDCDSC;SCSDCDSCS

Contents

1 Introduction **3**
1.1 Background . . . 3
1.2 Literature Review . . . 3
1.3 The Description of the Problem . . . 3
1.4 Our work . . . 4

2 title **4**
2.1 title . . . 4
2.1.1 title . . . 4

3 Assumptions **5**

4 Notations **6**
4.1 Proof of Lagrange Mean Value Theorem . . . 7
4.1.1 Step 1: Construct the Auxiliary Function . . . 7

5 Models **8**
5.1 Modeling and Solving of Problem 1 . . . 8
5.1.1 Problem Analysis . . . 8
5.1.2 Model Preparation . . . 8
5.1.3 Model Construction . . . 8
5.1.4 Model Solution . . . 8
5.2 Modeling and Solving of Problem 1 . . . 8
5.2.1 Problem Analysis . . . 8
5.2.2 Model Preparation . . . 8
5.2.3 Model Construction . . . 8
5.2.4 Model Solution . . . 8
5.3 Modeling and Solving of Problem 1 . . . 8
5.3.1 Problem Analysis . . . 8

5.3.2 Model Preparation 8
5.3.3 Model Construction 8
5.3.4 Model Solution 8

6 Sensitivity Analysis **9**

7 Model Evaluation and Promotion **10**
7.1 Model Evaluation 10
7.1.1 Advantages 10
7.1.2 Limitations 10
7.2 Future Work 10
7.2.1 Model extension 10
7.2.2 Model application 10

8 Conclusions **10**

1 Introduction

1.1 Background

good

good

<u>good</u>

good

good

1.2 Literature Review

good

good

good

good

good

good

good

good

good

1.3 The Description of the Problem

The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law. The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.
The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

1.4 Our work

2 title

2.1 title

2.1.1 title

1.title

3 Assumptions

Assumption 1: The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

Assumption 1:

Assumption 1:

Assumption 1: The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

Assumption 1: The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

- good
- good
  - good
  - good
    - good
    - good
      - good
      - good

1. good
2. good
   (a) good
   (b) good
      i. good
      ii. good
         A. good
         B. good

4 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: **Notations used in this literature**

Symbol	**Description**
PSNR	Peak signal-to-noise ratio for image quality evaluation
UCIQE	Underwater Color Image Quality Evaluation metric
UIQM	Underwater Image Quality Measure

Table 2: table

A	B	C
1	2	3
4	5	6

The core symbols and their definitions used in this study are summarized in Table2, providing an overview of the key parameters and their related meanings.

Table 3: table

Symbol	Description	C
PSNR	Peak signal-to-noise ratio for image quality evaluation	C

Figure 1: fig

4.1 Proof of Lagrange Mean Value Theorem

4.1.1 Step 1: Construct the Auxiliary Function

To apply Rolle's Theorem, we construct an auxiliary function $F(x)$, whose geometric meaning is "the difference between the ordinate of the curve $y = f(x)$ and the ordinate of the chord AB" (where $A(a, f(a))$ and $B(b, f(b))$ are the endpoints of the curve on $[a, b]$).

The equation of the straight line passing through A and B is:

$$y - f(a) = \frac{f(b) - f(a)}{b - a}(x - a)$$

which can be rewritten as:

$$y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$$

Thus, the auxiliary function is defined as:

$$F(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a}(x - a)\right]$$

To apply Rolle's Theorem, we construct an auxiliary function $F(x)$, whose geometric meaning is "the difference between

$$F(x)$$

$$F(x)$$

$$y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \tag{1}$$

5 Models

5.1 Modeling and Solving of Problem 1

5.1.1 Problem Analysis

5.1.2 Model Preparation

5.1.3 Model Construction

5.1.4 Model Solution

5.2 Modeling and Solving of Problem 1

5.2.1 Problem Analysis

5.2.2 Model Preparation

5.2.3 Model Construction

5.2.4 Model Solution

5.3 Modeling and Solving of Problem 1

5.3.1 Problem Analysis

5.3.2 Model Preparation

5.3.3 Model Construction

5.3.4 Model Solution

6 Sensitivity Analysis

7 Model Evaluation and Promotion

7.1 Model Evaluation

7.1.1 Advantages

7.1.2 Limitations

7.2 Future Work

7.2.1 Model extension

7.2.2 Model application

8 Conclusions

[2]

References

[1] Bundela B, Sharma S, Singh B R. A Review Article on Relation between Mathematical Modelling and Machine Learning[J]. Journal of China University of Mining & Technology, 2024, 29(2): 123–129.

[2] Krawitz J, Schukajlow S, Yang X, et al. A Systematic Review of International Perspectives on Mathematical Modelling: Modelling Goals and Task Characteristics[J]. ZDM – Mathematics Education, 2025, 57: 193–212.

[3] Lyon J A, Magana A J. A Review of Mathematical Modeling in Engineering Education[J]. International Journal of Engineering Education, 2020, 36(1A): 101–116.

[4] Huang S, Feng W, Tang C, et al. Partial Differential Equations Meet Deep Neural Networks: A Survey[EB/OL]. arXiv:2211.05567, 2022.

[5] Zino L, Cao M. Analysis, Prediction, and Control of Epidemics: A Survey from Scalar to Dynamic Network Models[EB/OL]. arXiv:2103.00181, 2021.

[6] Author Unknown. Review of Mathematical Modeling Research: A Descriptive Content Analysis[J]. Balıkesir NEF Journal, 2023.

[7] Author Unknown. Mathematical modelling, analysis and control in soft robotics: a survey[J]. Journal of the Brazilian Society of Mechanical Sciences and Engineering, 2023.

[8] Author Unknown. Mathematical Modelling at a Glance: A Theoretical Study[J]. Procedia - Social and Behavioral Sciences, 2012, 46: 3980–3984.

[9] Author Unknown. A Review of Recent Advances in Surrogate Models for Uncertainty Quantification[J]. Computer Methods in Applied Mechanics and Engineering, 2024.

Appendices

Report on Use of AI

1. Bing AI
   Query1:
   Output:

2. Bing AI
   Query1:
   Output:

3. Bing AI
   Query1:
   Output:

4. Bing AI
   Query1:
   Output: